

Markov Chain

1. Markov Process

A Markov process is a stochastic (random) process in which the future state depends only on the present state, not on the complete past history.

This property is called the Markov property.

If X_t represents the state of a system at time t , then

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots, X_0) = P(X_{t+1} = j \mid X_t = i).$$

Simple meaning

The future depends on the present, not on the past.

2. Markov Chain

A Markov chain is a particular type of Markov process in which the process usually has discrete time steps and, in elementary applications, a discrete set of states.

For example, suppose a computer server can be in one of three states:

$$S = \{\text{Normal, Busy, Failed}\}.$$

At every minute, the server changes from its current state to another state according to specified probabilities.

For example,

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0 & 0 & 1 \end{bmatrix}.$$

This is a Markov chain because we observe the system at discrete times:

$$t = 0, 1, 2, 3, \dots$$

and the next state depends only on the current state.

Easy relationship

Think of it this way:

$$\text{Markov Chain} \subset \text{Markov Processes}$$

That is,

Every Markov chain is a Markov process, but not every Markov process is necessarily a Markov chain.

3. Main Difference

Feature	Markov Process	Markov Chain
Meaning	General class of stochastic processes satisfying the Markov property	A particular type of Markov process
Time	Can be discrete or continuous	Usually, discrete time
State space	Can be discrete or continuous	Often discrete in elementary Markov-chain courses
Transition matrix	Not always represented by a simple matrix	Usually represented by a transition probability matrix
Example	Continuous-time customer arrival process	Web-page navigation at each click

Why Markov Chains are Useful in Computer Science

Markov chains are used in:

1. Web page ranking and navigation analysis
2. Computer network traffic modeling
3. Queueing and server performance analysis
4. Reliability of computer systems
5. Natural Language Processing (NLP)
6. Speech recognition
7. Text generation
8. User behavior prediction
9. Machine learning and AI

Markov Chain

Definition: A sequence of $[X_n]$ is a Markov chain if each X_n is a random variable and if, for all n ,

$$P[X_n = x_n | X_{n-1} = x_{n-1}, \dots | X_0 = x_0] = P[X_n = x_n | X_{n-1} = x_{n-1}]$$

Then the process $[X_n; n = 0, 1, 2, \dots]$ is called a Markov chain.

Here, $x_1, x_2, \dots, x_n, \dots$ are called the states (outcomes) of the Markov chain.

In a Markovian process with n exhaustive and mutually exclusive states (outcomes) the probabilities at a specific point in time $t = 0, 1, 2, \dots$ is usually written as

$$P_{ij} = P\{X(t) = j | X(t-1) = i\}, (i, j) = 1, 2, \dots, n, \quad t = 0, 1, 2, \dots$$

This is known as the one step transition probability of moving from state i at $t-1$, to state j at t .

OR

$[P_{ij}(n-1, n) = P(X_n = x_j | X_{n-1} = x_i)]$ is called the one step transition probability from state x_i to state x_j in the n th step.]

If the one step transition probability does not depend on the step i.e., independent of n . then the Markov chain is called a homogeneous Markov chain.

If the Markov chain is homogeneous, the one step transition probability is denoted by P_{ij} and the matrix $P = [P_{ij}]$ is called the one step transition probability matrix (TPM)

$$\text{i.e., } P = \begin{bmatrix} P_{11} & P_{12} & P_{13} & \dots & P_{1n} \\ P_{21} & P_{22} & P_{23} & \dots & P_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ P_{n1} & P_{n2} & P_{n3} & \dots & P_{nn} \end{bmatrix}$$

The matrix P defines the so-called Markov chain. It has the property that all its transition probabilities P_{ij} are fixed and independent over time.

Definitions:

1. The conditional probability $P_{ij}^{(n)} = P(X_n = x_j | x_0 = x_i)$ is called the n -step transition probability.
2. The P_i = Probability that the process is in state x_i at any step ($i = 1, 2, \dots, k$). Then the row vector $p = (p_1, p_2, \dots, p_k)$ is called the probability distribution of the process at that time.
3. $p^{(0)} = (p_1^{(0)}, p_2^{(0)}, \dots, p_k^{(0)})$ is called the initial probability distribution. Where
 $p_1^{(0)} = P(X_0 = 1), p_2^{(0)} = P(X_0 = 2), \dots$ are the initial probabilities for states 1, 2,

CHAPMAN - KOLMOGOROV EQUATION

The **Chapman–Kolmogorov equation** is one of the fundamental equations in the theory of **Markov processes and Markov chains**.

It tells us how to calculate the probability of moving from one state to another in **several steps** by considering all possible intermediate states.

1. Basic Idea

Suppose a Markov chain has three states:

$$A, \quad B, \quad C.$$

Suppose we want to find the probability of going from A to C in 2 steps.

There may be different paths:

$$A \rightarrow A \rightarrow C$$

$$A \rightarrow B \rightarrow C$$

$$A \rightarrow C \rightarrow C.$$

Therefore, we calculate the probability of each possible intermediate state and add them.

This is the basic idea behind the **Chapman–Kolmogorov equation**.

2. Mathematical Form

Let

$$P_{ij}^{(n)}$$

denote the probability of going from state i to state j in n steps.

Then the Chapman–Kolmogorov equation is

$$P_{ij}^{(m+n)} = \sum_k P_{ik}^{(m)} P_{kj}^{(n)}$$

where the summation is over all possible intermediate states k .

3. Interpretation

The equation says:

To find the probability of going from state i to state j in $m + n$ steps, consider every possible state k that the process can occupy after m steps.

For each intermediate state k :

$$\text{Probability of } i \rightarrow k \times \text{Probability of } k \rightarrow j$$

Then add these probabilities for all possible k .

4. Simple Numerical Example

Consider the transition matrix

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

with states A and B .

Suppose we want to find

$$P_{AA}^{(2)},$$

the probability of starting at A and being back at A after 2 steps.

There are two possible intermediate states.

Path 1: $A \rightarrow A \rightarrow A$

$$\begin{aligned} &P(A \rightarrow A)P(A \rightarrow A) \\ &= 0.6 \times 0.6 = 0.36. \end{aligned}$$

Path 2: $A \rightarrow B \rightarrow A$

$$\begin{aligned} &P(A \rightarrow B)P(B \rightarrow A) \\ &= 0.4 \times 0.3 = 0.12. \end{aligned}$$

Therefore,

$$\begin{aligned} P_{AA}^{(2)} &= 0.36 + 0.12 \\ &= 0.48. \end{aligned}$$

5. Chapman–Kolmogorov Equation for This Example

Using the formula,

$$P_{AA}^{(2)} = \sum_k P_{Ak}^{(1)} P_{kA}^{(1)}.$$

Since the intermediate state k can be A or B ,

$$P_{AA}^{(2)} = P_{AA}^{(1)} P_{AA}^{(1)} + P_{AB}^{(1)} P_{BA}^{(1)}.$$

Therefore,

$$\begin{aligned} &= 0.6(0.6) + 0.4(0.3) \\ &= 0.48. \end{aligned}$$

6. Matrix Form of Chapman–Kolmogorov Equation

The Chapman–Kolmogorov equation has a very important matrix interpretation.

For a time-homogeneous Markov chain,

$$P^{m+n} = P^m P^n$$

where P^m and P^n are the m -step and n -step transition matrices.

In particular,

$$P^{(2)} = P \times P = P^2$$

and

$$P^{(3)} = P^2 P = P^3.$$

More generally,

$$P^{(n)} = P^n.$$

This is why matrix multiplication is so important in Markov chains.

Simple Three-State Example

Suppose a student can be in one of three states:

- A = Classroom
- B = Library
- C = Canteen

Assume the transition probability matrix is

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}.$$

For example:

- From A , the student stays in A with probability $1/2$ and goes to B with probability $1/2$.
- From B , the student stays in B with probability $1/2$ and goes to C with probability $1/2$.
- From C , the student goes to A with probability $1/2$ and stays in C with probability $1/2$.

Chapman–Kolmogorov Equation

Suppose we want to find:

What is the probability that a student currently in A will be in C after 2 transitions?

We write

$$P_{AC}^{(2)}.$$

The student can reach C after two steps through different intermediate states.

Since there are three states, the intermediate state can be A , B , or C .

Therefore,

$$P_{AC}^{(2)} = P_{AA}P_{AC} + P_{AB}P_{BC} + P_{AC}P_{CC}.$$

This is the Chapman–Kolmogorov equation.

Step 1: Path through A

$$A \rightarrow A \rightarrow C$$

But

$$P(A \rightarrow C) = 0.$$

Therefore,

$$P_{AA}P_{AC} = \frac{1}{2}(0) = 0.$$

Step 2: Path through B

$$A \rightarrow B \rightarrow C$$

We have

$$P(A \rightarrow B) = \frac{1}{2}$$

and

$$P(B \rightarrow C) = \frac{1}{2}.$$

Therefore,

$$P_{AB}P_{BC} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Step 3: Path through C

$$A \rightarrow C \rightarrow C$$

But

$$P(A \rightarrow C) = 0.$$

Therefore,

$$P_{AC}P_{CC} = 0 \times \frac{1}{2} = 0.$$

Final Calculation

Therefore,

$$P_{AC}^{(2)} = 0 + \frac{1}{4} + 0$$

$$\boxed{P_{AC}^{(2)} = \frac{1}{4}}$$

So there is a **25% probability** that a student starting in the Classroom will be in the Canteen after two transitions.

Verify Using Matrix Multiplication

The Chapman–Kolmogorov equation can also be obtained from

$$P^2 = P \times P.$$

Here,

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}.$$

Therefore,

$$P^2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}.$$

The (1, 3) element is

$$\begin{aligned} (P^2)_{13} &= \frac{1}{2}(0) + \frac{1}{2}\left(\frac{1}{2}\right) + 0\left(\frac{1}{2}\right) \\ &= \frac{1}{4}. \end{aligned}$$

This is exactly what we obtained using the Chapman–Kolmogorov equation.

For three states, remember:

$$P_{ij}^{(2)} = P_{iA}P_{Aj} + P_{iB}P_{Bj} + P_{iC}P_{Cj}$$

In words:

Start at the initial state → consider every possible intermediate state → reach the final state → multiply each path probability → add all the path probabilities.

For our example:

$$A \rightarrow C \text{ in 2 steps}$$

can be calculated as

$$\begin{aligned} &(A \rightarrow A \rightarrow C) + (A \rightarrow B \rightarrow C) + (A \rightarrow C \rightarrow C) \\ &= 0 + \frac{1}{4} + 0 = \frac{1}{4}. \end{aligned}$$

Classification of States and Markov Chains

In a **Markov chain**, the system moves from one state to another according to transition probabilities. The states and the chain itself can be classified based on how the states communicate and how the process behaves over time.

1. Classification of States

Consider a Markov chain with transition matrix $P = (p_{ij})$, where

$$p_{ij} = P(X_{n+1} = j \mid X_n = i).$$

The main classifications of states are:

1. Accessible state
2. Communicating states
3. Transient state
4. Recurrent state
5. Absorbing state
6. Periodic state
7. Aperiodic state
8. Closed state
9. Essential and inessential states

1. Accessible State

Definition

State j is **accessible** from state i if there is some number of steps $n \geq 1$ such that

$$p_{ij}^{(n)} > 0.$$

Here, $p_{ij}^{(n)}$ is the probability of reaching j from i in exactly n steps.

We write

$$i \rightarrow j.$$

Example

Suppose

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}.$$

From state 1:

$$1 \rightarrow 2$$

because $p_{12} = 0.5 > 0$.

Also,

$$1 \rightarrow 3$$

because the chain can go

$$1 \rightarrow 2 \rightarrow 3.$$

Therefore, state 3 is accessible from state 1.

2. Communicating States

Definition

Two states i and j are said to **communicate** if each is accessible from the other.

That is,

$$i \leftrightarrow j$$

if

$$i \rightarrow j \quad \text{and} \quad j \rightarrow i.$$

Example

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

States 1 and 2 communicate because

$$1 \rightarrow 2$$

and

$$2 \rightarrow 1.$$

Therefore,

$$1 \leftrightarrow 2.$$

But state 3 does not communicate with states 1 or 2 because there is no path from 3 back to them.

Communication Class

A group of states that communicate with each other is called a communicating class.

In this example:

$$\{1, 2\}$$

is one communicating class, and

$$\{3\}$$

is another.

3. Transient State

Definition

A state is called **transient** if, after leaving that state, there is a positive probability that the process will never return to it.

In simple words:

Transient = a state that may be visited only temporarily.

Example

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}.$$

From state 1, the process can move to state 2 and eventually to state 3.

Once it reaches state 3, it remains there.

Therefore, after leaving state 1, it cannot come back to state 1.

Hence,

State 1 is transient.

Similarly, state 2 is also transient.

4. Recurrent State

Definition

A state is **recurrent** if, starting from that state, the probability of eventually returning to the same state is 1.

In simple words:

Recurrent = the process will return to the state with certainty.

Example

Consider

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The movement is

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 1 \dots$$

Starting from state 1, we always return to state 1.

Therefore, state 1 is recurrent.

Similarly, state 2 is recurrent.

States 1 and 2 are recurrent.

5. Absorbing State

Definition

A state i is called an **absorbing state** if, once the process enters that state, it remains there permanently.

Mathematically,

$$p_{ii} = 1.$$

Therefore,

$$p_{ij} = 0, \quad j \neq i.$$

Example

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}.$$

For state 3,

$$p_{33} = 1.$$

Thus,

$$3 \rightarrow 3 \rightarrow 3 \rightarrow 3 \dots$$

Therefore,

State 3 is absorbing.

Real-life example

Consider a machine:

- State 1 = Working
- State 2 = Under repair
- State 3 = Permanently failed

If a permanently failed machine cannot return to operation, state 3 is an absorbing state.

6. Periodic State

Definition

A state i has **period** d if the process can return to state i only at multiples of d steps, where d is the greatest common divisor of all possible return times.

Mathematically,

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

If

$$d(i) > 1,$$

the state is **periodic**.

Example

Consider

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Starting from state 1:

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 1.$$

We can return to state 1 only after

$$2, 4, 6, 8, \dots$$

steps.

Therefore,

$$d(1) = \gcd(2, 4, 6, \dots) = 2.$$

Hence, state 1 is periodic with period 2.

State 2 is also periodic with period 2.

7. Aperiodic State

Definition

A state is **aperiodic** if its period is

$$d = 1.$$

In other words, there is no fixed cycle length greater than 1 restricting the possible return times.

Example

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

Starting from state 1, it is possible to return to state 1 in one step because

$$p_{11} = 0.5 > 0.$$

Since the possible return times include 1,

$$d(1) = 1.$$

Therefore,

State 1 is aperiodic.

State 2 is also aperiodic.

8. Closed State / Closed Class

Definition

A set of states is called a **closed class** if, once the Markov chain enters the class, it cannot leave the class.

Example

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The set

$$\{1, 2\}$$

is closed because transitions from states 1 and 2 never lead to state 3.

Similarly,

$$\{3\}$$

is also a closed class.

9. Essential and Inessential States

Essential State

A state is **essential** if it belongs to a closed communicating class.

Once the chain enters an essential class, it cannot leave that class.

Inessential State

A state is **inessential** if it belongs to a class from which the chain can move to another class and cannot necessarily return.

Example

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}.$$

The classes are

$$\{1\}, \quad \{2\}, \quad \{3\}.$$

State 3 is essential because it is a closed class.

States 1 and 2 are inessential because the process can eventually move toward state 3 and cannot return.

Classification Summary

Type of state	Simple meaning	Example
Accessible	Can reach another state	$1 \rightarrow 2$
Communicating	Two states can reach each other	$1 \leftrightarrow 2$
Transient	May leave and never return	States 1, 2 in absorbing-chain example
Recurrent	Returns with probability 1	States in $1 \leftrightarrow 2$ chain
Absorbing	Once entered, never leaves	$p_{33} = 1$
Periodic	Returns only at multiples of $d > 1$	Two-state alternating chain, $d = 2$
Aperiodic	Period is 1	Chain with $p_{ii} > 0$
Closed	Cannot leave the class	$\{1, 2\}$ in the example
Essential	Belongs to a closed class	State 3
Inessential	Can move to another class	States 1 and 2

10. Classification of Markov Chains

Markov chains themselves can also be classified.

A. Irreducible Markov Chain

Definition

A Markov chain is **irreducible** if every state can be reached from every other state.

That is,

$$i \leftrightarrow j$$

for every pair of states i, j .

Example

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.2 & 0.3 & 0.5 \\ 0.4 & 0 & 0.6 \end{bmatrix}.$$

There is a path between every pair of states.

Therefore, the chain is irreducible.

Simple meaning:

Every state communicates with every other state.

11. Reducible Markov Chain

Definition

A Markov chain is **reducible** if at least one state cannot be reached from another state.

Example

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}.$$

State 3 can be reached from state 1, but state 1 cannot be reached from state 3.

Therefore, the chain is reducible.

12. Periodic Markov Chain

A Markov chain is **periodic** if its states have a common period greater than 1.

Example

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The chain alternates:

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow \dots$$

Both states have period 2.

Therefore, the chain is periodic with period 2.

13. Aperiodic Markov Chain

A Markov chain is **aperiodic** if its states have period 1.

Example

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

Because the process can remain in the same state in one step,

$$p_{11} > 0, \quad p_{22} > 0,$$

the states have period 1.

Hence, the chain is aperiodic.

14. Absorbing Markov Chain

Definition

A Markov chain is an **absorbing Markov chain** if:

1. It has at least one absorbing state, and
2. Every state can reach an absorbing state.

Example

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}.$$

State 3 is absorbing:

$$p_{33} = 1.$$

States 1 and 2 can eventually reach state 3.

Therefore, this is an absorbing Markov chain.

15. Ergodic Markov Chain

A Markov chain is commonly called **ergodic** when it is both:

- irreducible, and
- aperiodic,

with the usual finite-state setting giving a unique long-run stationary distribution.

Example

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}.$$

Both states communicate, so the chain is irreducible.

Also,

$$p_{11} = 0.7 > 0, \quad p_{22} = 0.6 > 0,$$

so it is aperiodic.

Hence, this is an **ergodic Markov chain**.

In a Markov chain, the terms **persistent** and **non-persistent** (more commonly **transient**) describe what happens when the chain visits a state.

1. Persistent State

A state i is called **persistent (recurrent)** if, after starting from state i , the Markov chain will **eventually return to i** with probability 1.

Mathematically,

$$P_i(\text{ever return to } i) = 1$$

Equivalently, if f_{ii} is the probability of ever returning to state i ,

$$\boxed{f_{ii} = 1}$$

Example

Consider the Markov chain:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The chain moves:

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow \dots$$

Starting from state 1, it **always** returns to state 1.

Therefore,

$$\boxed{\text{State 1 is persistent}}$$

Similarly, state 2 is also persistent.

2. Non-Persistent State

A state is called non-persistent, or transient, if there is a positive probability that the chain will never return to that state after leaving it.

Thus,

$$P_i(\text{ever return to } i) < 1$$

or

$f_{ii} < 1$

Example

Consider

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0 & 1 \end{bmatrix}$$

Here:

- From state 1, there is a 0.5 probability of going to state 2.
- Once the chain reaches state 2, it stays there forever.

So, starting from state 1, there is a positive probability that the chain goes:

$$1 \rightarrow 2 \rightarrow 2 \rightarrow 2 \rightarrow \dots$$

and never returns to state 1.

Therefore,

State 1 is non-persistent (transient)

State 2 is persistent because once it is reached, the chain remains there.

3. Persistent vs Non-Persistent

Feature	Persistent (Recurrent)	Non-Persistent (Transient)
Meaning	State will eventually be revisited	State may never be revisited
Return probability	$f_{ii} = 1$	$f_{ii} < 1$
Probability of never returning	0	> 0
Number of visits	Infinitely many visits with probability 1	May be visited only finitely many times
Common term	Recurrent	Transient

4. An important further classification

Persistent states can be further divided into:

(a) Null Persistent / Null Recurrent

A persistent state i is null persistent if the mean recurrence time is infinite.

$$E_i(T_i) = \infty$$

where T_i is the time taken to return to state i .

So the chain is guaranteed to return eventually, but the expected waiting time for the return is infinite.

(b) Non-Null Persistent / Positive Recurrent

A persistent state is non-null persistent (also called positive recurrent) if the mean recurrence time is finite:

$$E_i(T_i) < \infty$$

For an irreducible positive recurrent Markov chain, the stationary probability is related to the mean recurrence time by

$$\pi_i = \frac{1}{E_i(T_i)}$$

Complete classification

$$\text{Markov State}$$

$$\begin{cases} \text{Transient (Non-persistent)} & f_{ii} < 1 \\ \text{Persistent (Recurrent)} & f_{ii} = 1 \end{cases}$$

and persistent states are further divided into

$$\begin{cases} \text{Null Persistent} & E_i(T_i) = \infty \\ \text{Non-null Persistent} & E_i(T_i) < \infty \end{cases}$$

In short:

Non-persistent $\rightarrow$ may never return.

Null persistent $\rightarrow$ must return, but expected return time is infinite.

Non-null persistent $\rightarrow$ must return, and expected return time is finite.

Suppose a computer server has three states:

- 0 = Low load
- 1 = Medium load
- 2 = High load

Consider

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

The server can move from any state to another state.

For example:

$$P(0 \rightarrow 1) = 0.3$$

means that if the server has low load today, there is a 30% chance of medium load tomorrow.

Because this is a finite irreducible Markov chain, every state is positive recurrent (non-null persistent).

Therefore:

$$0, 1, 2 \text{ are all non-null persistent states}$$

and each state has a finite expected return time.

1. Transient / Non-persistent

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow \dots$$

There is a possibility that the chain never returns to 1.

$$P(\text{return to } 1) < 1$$

2. Null persistent

The chain is certain to return to 1, but the expected time for returning is infinite:

$$P(\text{return}) = 1$$

but

$$E(T_1) = \infty$$

3. Non-null persistent / Positive recurrent

The chain is **certain to return**, and the expected return time is finite:

$$P(\text{return}) = 1$$

and

$$E(T_1) < \infty$$

Easy memory trick

Persistent = "I will come back."

Null persistent = "I will come back, but I don't know when—the expected wait is infinite."

Non-null persistent = "I will come back, and the expected waiting time is finite."

For finite-state Markov chains, an especially important result is:

Finite + irreducible Markov chain $\Rightarrow$ every state is non-null persistent (positive recurrent).

Problem-1:

Let us take a 4×4 Transition probability matrix, check for **irreducibility, periodicity, aperiodicity, and ergodicity**.

Consider

$$P = \begin{pmatrix} 0.4 & 0.6 & 0 & 0 \\ 0.2 & 0.3 & 0.5 & 0 \\ 0 & 0.4 & 0.2 & 0.4 \\ 0.1 & 0 & 0.3 & 0.6 \end{pmatrix}.$$

Problem-2:

Let us take a 5×5 Transition probability matrix, check for **irreducibility, periodicity, aperiodicity, and ergodicity**.

Consider

$$P = \begin{bmatrix} 0.2 & 0.3 & 0.1 & 0.2 & 0.2 \\ 0.1 & 0.3 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.1 & 0.3 & 0.2 & 0.2 \\ 0.1 & 0.2 & 0.2 & 0.3 & 0.2 \\ 0.2 & 0.2 & 0.1 & 0.2 & 0.3 \end{bmatrix}$$

Let us take a **simple 3-state transition probability matrix (TPM)** so that we can clearly calculate **pure probabilities, conditional probabilities, and sequential probabilities**.

Example TPM

Suppose a system can be in three states:

- $S_1 = \text{Low}$
- $S_2 = \text{Medium}$
- $S_3 = \text{High}$

Let the transition probability matrix be

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

The rows represent the **current state** and columns represent the **next state**.

1. Pure Probabilities

A pure probability means the probability of a state at a particular time without specifying the previous state.

Suppose initially,

$$P(X_0 = S_1) = 0.4,$$

$$P(X_0 = S_2) = 0.4,$$

$$P(X_0 = S_3) = 0.2.$$

Therefore, the initial probability vector is

$$\pi^{(0)} = [0.4 \quad 0.4 \quad 0.2].$$

The pure probabilities at time 1 are obtained by

$$\boxed{\pi^{(1)} = \pi^{(0)} P}$$

Thus,

$$[0.4 \quad 0.4 \quad 0.2] \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

Therefore,

$$\begin{aligned} P(X_1 = S_1) &= 0.4(0.5) + 0.4(0.2) + 0.2(0.1) \\ &= 0.20 + 0.08 + 0.02 = \boxed{0.30} \end{aligned}$$

Similarly,

$$\begin{aligned} P(X_1 = S_2) &= 0.4(0.3) + 0.4(0.5) + 0.2(0.4) \\ &= 0.12 + 0.20 + 0.08 = \boxed{0.40} \end{aligned}$$

and

$$\begin{aligned} P(X_1 = S_3) &= 0.4(0.2) + 0.4(0.3) + 0.2(0.5) \\ &= 0.08 + 0.12 + 0.10 = \boxed{0.30} \end{aligned}$$

So,

$$\pi^{(1)} = [0.30, 0.40, 0.30]$$

These are pure probabilities.

2. Conditional Probabilities

A conditional probability tells us the probability of the **next state given the present state**.

These probabilities are directly available from the TPM.

For example,

$$P(X_{n+1} = S_2 \mid X_n = S_1) = p_{12}$$

Therefore,

$$P(X_{n+1} = S_2 \mid X_n = S_1) = 0.3$$

Similarly,

$$P(X_{n+1} = S_3 \mid X_n = S_2) = p_{23} = \boxed{0.3}$$

and

$$P(X_{n+1} = S_1 \mid X_n = S_3) = p_{31} = \boxed{0.1}.$$

In general

For a TPM,

$$p_{ij} = P(X_{n+1} = j \mid X_n = i)$$

So **each element of the TPM is a one-step conditional probability**.

3. Sequential Probabilities

This is particularly important in Markov chains.

A **sequential probability** is the probability of following a **sequence of states**.

Suppose we want:

$$S_1 \rightarrow S_2 \rightarrow S_3.$$

That means:

$$X_0 = S_1, \quad X_1 = S_2, \quad X_2 = S_3.$$

We have

$$P(S_2 \mid S_1) = 0.3$$

and

$$P(S_3 \mid S_2) = 0.3.$$

Therefore,

$$P(S_1 \rightarrow S_2 \rightarrow S_3)$$

is

$$P(S_1) \times P(S_2 \mid S_1) \times P(S_3 \mid S_2).$$

If

$$P(S_1) = 0.4,$$

then

$$= 0.4(0.3)(0.3)$$

$$= \boxed{0.036}.$$

So there is a **3.6% probability** of the complete sequence

$$\boxed{S_1 \rightarrow S_2 \rightarrow S_3}.$$

4. Another sequential probability

Suppose we want the sequence

$$S_2 \rightarrow S_3 \rightarrow S_1.$$

We know:

$$P(S_2) = 0.4$$

$$P(S_3 | S_2) = 0.3$$

$$P(S_1 | S_3) = 0.1.$$

Therefore,

$$\begin{aligned} P(S_2 \rightarrow S_3 \rightarrow S_1) \\ &= 0.4(0.3)(0.1) \\ &= \boxed{0.012}. \end{aligned}$$

Thus, the probability of this particular sequence is

$$\boxed{1.2\%}.$$

5. Difference between the three

Type	Meaning	Example
Pure probability	Probability of being in a state at a given time, without conditioning on previous state	$P(X_1 = S_2) = 0.40$
Conditional probability	Probability of next state given current state	$P(S_2 S_1) = 0.30$
Sequential probability	Probability of a complete sequence of states	$P(S_1 \rightarrow S_2 \rightarrow S_3) = 0.036$

Suppose the states represent **server load**:

$$L = \text{Low}, \quad M = \text{Medium}, \quad H = \text{High}.$$

- **Pure:** "What is the probability that tomorrow's load is Medium?"

$$P(X_1 = M) = 0.40$$

- **Conditional:** "If today's load is Low, what is the probability that tomorrow it is Medium?"

$$P(M | L) = 0.30$$

- **Sequential:** "What is the probability that the load follows Low $\rightarrow$ Medium $\rightarrow$ High?"

$$P(L \rightarrow M \rightarrow H) = 0.036$$

So remember:

$$\boxed{\text{Pure} = \text{state probability}}$$

$$\boxed{\text{Conditional} = \text{one transition probability}}$$

$$\boxed{\text{Sequential} = \text{probability of a path/sequence}}$$

Problem: A fair die is tossed repeatedly. If X_n , denotes the maximum of the numbers occurring in the first n tosses, find the transition probability matrix P of the Markov chain $\{X_n\}$. Find also P^2 and $P(X_2 = 6)$.

Solution:

Fair Die Tossed Repeatedly — Markov Chain

A fair die is tossed repeatedly. Let

$$X_n = \text{maximum number obtained in the first } n \text{ tosses.}$$

We need to find:

1. Transition probability matrix P
2. P^2
3. $P(X_2 = 6)$

1. Why is $\{X_n\}$ a Markov Chain?

At any stage, suppose the current maximum is i .

On the next toss:

- If the new number is $\leq i$, the maximum remains i .
- If the new number is $j > i$, the new maximum becomes j .

Therefore, the next value of X_n depends only on the current value $X_n = i$, not on the previous tosses.

Hence,

$$\boxed{\{X_n\} \text{ is a Markov chain.}}$$

The possible states are

$$\boxed{S = \{1, 2, 3, 4, 5, 6\}}.$$

2. Transition Probabilities

Suppose the current maximum is i .

Case 1: The maximum remains i

This happens when the next die toss is

$$1, 2, \dots, i.$$

There are i such outcomes out of 6.

Therefore,

$$P_{ii} = P(X_{n+1} = i \mid X_n = i) = \frac{i}{6}$$

for $i = 1, 2, \dots, 5$.

For example, if the current maximum is 3, the next maximum remains 3 if the next toss is 1, 2, or 3.

Thus,

$$P_{33} = \frac{3}{6} = \frac{1}{2}.$$

Case 2: Maximum increases from i to j

If

$$j > i,$$

then the next toss must be exactly j .

Since the die is fair,

$$P_{ij} = \frac{1}{6}, \quad j > i.$$

For example, if the current maximum is 3,

$$P(3 \rightarrow 4) = \frac{1}{6},$$

$$P(3 \rightarrow 5) = \frac{1}{6},$$

$$P(3 \rightarrow 6) = \frac{1}{6}.$$

Case 3: Maximum is 6

Once the maximum reaches 6, it can never decrease.

Therefore,

$$P_{66} = 1.$$

Transition probabilities

$$P_{ij} = \begin{cases} \frac{i}{6}, & j = i, \quad i = 1, \dots, 5, \\ \frac{1}{6}, & j > i, \\ 0, & j < i, \\ 1, & i = j = 6. \end{cases}$$

3. Transition Probability Matrix P

Using the state order

1, 2, 3, 4, 5, 6,

we get

$$P = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{3}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Each row sums to 1.

For example, the third row is

$$0 + 0 + \frac{1}{2} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1.$$

4. Finding P^2

The two-step transition matrix is

$$P^2 = P \times P.$$

Here P_{ij}^2 represents

$$P(X_{n+2} = j \mid X_n = i).$$

A useful observation makes the multiplication easier.

If $j > i$

To have maximum j after two additional tosses, at least one of those two tosses must be j , while neither can exceed j .

Therefore,

$$P_{ij}^2 = \left(\frac{j}{6}\right)^2 - \left(\frac{j-1}{6}\right)^2.$$

Hence,

$$P_{ij}^2 = \frac{j^2 - (j-1)^2}{36} = \boxed{\frac{2j-1}{36}}.$$

For $j = i$, both tosses must be $\leq i$, so

$$\boxed{P_{ii}^2 = \frac{i^2}{36}}.$$

Thus,

$$P^2 = \begin{bmatrix} \frac{1}{36} & \frac{3}{36} & \frac{5}{36} & \frac{7}{36} & \frac{9}{36} & \frac{11}{36} \\ 0 & \frac{4}{36} & \frac{5}{36} & \frac{7}{36} & \frac{9}{36} & \frac{11}{36} \\ 0 & 0 & \frac{9}{36} & \frac{7}{36} & \frac{9}{36} & \frac{11}{36} \\ 0 & 0 & 0 & \frac{16}{36} & \frac{9}{36} & \frac{11}{36} \\ 0 & 0 & 0 & 0 & \frac{25}{36} & \frac{11}{36} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

5. Find $P(X_2 = 6)$

Since

$X_2 =$ maximum of the first two tosses,

$X_2 = 6$ means that at least one of the two tosses is 6.

There are two ways to calculate it.

Method 1: Complement

The probability that the first two tosses are not 6 is

$$\left(\frac{5}{6}\right)^2 = \frac{25}{36}.$$

Therefore,

$$\begin{aligned} P(X_2 = 6) &= 1 - \frac{25}{36} \\ &= \frac{11}{36}. \end{aligned}$$

$$\begin{aligned} P(X_2 = 6) &= \sum_{i=1}^6 P(X_1 = i) \times P_{i6}^1 \\ &= \frac{1}{6} \times \frac{1}{6} \times (1 + 1 + 1 + 1 + 1 + 6) = \frac{11}{36} \end{aligned}$$

Method 2: Using P^2

Initially, after zero tosses, the maximum is not naturally one of the states 1, 2,6. But after the first toss, the distribution of X_1 is

$$P(X_1 = i) = \frac{1}{6}, \quad i = 1, \dots, 6$$

Therefore

$$\begin{aligned} P(X_2 = 6) &= \sum_{i=1}^6 P(X_1 = i) \times P_{i6}^2 \\ &= \frac{1}{6} \times \frac{1}{36} \times (11 + 11 + 11 + 11 + 11 + 36) = \frac{91}{216} \end{aligned}$$

Problem: An urn initially contains five black balls and five white balls. The following experiment is repeated indefinitely. A ball is drawn from the urn; if the ball is white, it is put back in the urn, otherwise it is left out. Let X_n be the number of black balls remaining in the urn after n draws from the urn.

Is X_n a Markov process? If so,

(a) Find the appropriate transition probabilities.

(b) Find the one - step transition probability matrix P for X_n .

(c) Find the two-step transition probability matrix P^2 by matrix multiplication.

(d) What happens to X_n , as n approaches infinity? Use your answer to guess the limit of P^n as $n \rightarrow \infty$.

Solution: Given that

An urn initially contains:

- 5 black balls
- 5 white balls

At each draw:

- If a white ball is drawn, it is replaced.
- If a black ball is drawn, it is not replaced.

Let X_n = number of black balls remaining after n draws.

The possible states are

$$S = \{0, 1, 2, 3, 4, 5\}.$$

Initially,

$$X_0 = 5.$$

Is $\{X_n\}$ a Markov process?

Yes.

The probability of the next state depends only on the current number of black balls, not on the previous history.

If $X_n = i$, then there are:

- i black balls
- 5 white balls
- $i + 5$ total balls.

Therefore,

$$P(X_{n+1} = i - 1 \mid X_n = i) = \frac{i}{i + 5},$$

and

$$P(X_{n+1} = i \mid X_n = i) = \frac{5}{i + 5}.$$

Thus, $\{X_n\}$ is a Markov chain.

(a) Appropriate transition probabilities

For state i , $i = 1, 2, 3, 4, 5$:

Transition to $i - 1$

A black ball is selected:

$$p_{i,i-1} = \frac{i}{i+5}$$

Transition to i

A white ball is selected and replaced:

$$p_{i,i} = \frac{5}{i+5}$$

For state 0 , there are no black balls left, so it remains at 0 :

$$p_{00} = 1.$$

Hence,

$$p_{ij} = 0$$

for all other transitions.

For example, if $X_n = 3$,

$$P(3 \rightarrow 2) = \frac{3}{8}, \quad P(3 \rightarrow 3) = \frac{5}{8}.$$

(b) One-step transition probability matrix P

Take the state order as

$$0, 1, 2, 3, 4, 5.$$

The transition matrix is

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{6} & \frac{5}{6} & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{7} & \frac{5}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{8} & \frac{5}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{4}{9} & \frac{5}{9} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Notice that the matrix is **lower triangular** because the number of black balls can either remain the same or decrease by one.

(c) Find P^2 by matrix multiplication

We calculate

$$P^2 = P \times P.$$

For example,

$$\begin{aligned}(P^2)_{1,0} &= p_{1,0}p_{0,0} + p_{1,1}p_{1,0} \\ &= \frac{1}{6}(1) + \frac{5}{6}\left(\frac{1}{6}\right) = \frac{1}{6} + \frac{5}{36} = \frac{11}{36}.\end{aligned}$$

Similarly,

$$(P^2)_{1,1} = \frac{5}{6} \cdot \frac{5}{6} = \frac{25}{36}.$$

Therefore,

$$P^2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{11}{36} & \frac{25}{36} & 0 & 0 & 0 & 0 \\ \frac{1}{21} & \frac{65}{147} & \frac{25}{49} & 0 & 0 & 0 \\ 0 & \frac{3}{28} & \frac{225}{448} & \frac{25}{64} & 0 & 0 \\ 0 & 0 & \frac{1}{6} & \frac{85}{162} & \frac{25}{81} & 0 \\ 0 & 0 & 0 & \frac{2}{9} & \frac{19}{36} & \frac{1}{4} \end{pmatrix}$$

The matrix multiplication follows the row-column rule:

(d) What happens to X_n as $n \rightarrow \infty$?

This is the important part.

Whenever a black ball is drawn, it is permanently removed.

Therefore, the number of black balls can only:

$$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0.$$

It can never increase.

Eventually, **all five black balls will be removed with probability 1.**

Hence,

$$X_n \longrightarrow 0 \quad \text{as } n \rightarrow \infty$$

with probability 1.

State 0 is an **absorbing state**, because

$$P(0 \rightarrow 0) = 1.$$

Limit of P^n

Since every state eventually reaches state 0, the long-run transition probabilities satisfy

$$\lim_{n \rightarrow \infty} P^n = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus,

$$\lim_{n \rightarrow \infty} P^n = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Long-run interpretation

No matter how many black balls are initially present, eventually:

$$\text{Number of black balls remaining} = 0$$

with probability 1.

Since the experiment starts with $X_0 = 5$, the fifth row of P^n gives the distribution of X_n . Therefore,

$$\lim_{n \rightarrow \infty} P(X_n = 0) = 1$$

and

$$\lim_{n \rightarrow \infty} P(X_n = j) = 0, \quad j = 1, 2, 3, 4, 5.$$

Problem: The three state Markov chain is given by the transition probability matrix

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \text{ Prove that the chain is irreducible.}$$

Answer: the given chain is irreducible and all the states are recurrent.

Problem: The three state Markov chain is given by the transition probability matrix

$$P = \begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix} \text{ Is this matrix irreducible?}$$

Answer: The given matrix is irreducible and all the states are recurrent.

Problem: Consider the Markov chain with transition probability matrix

$$P = \begin{bmatrix} 0.4 & 0.6 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ Is this Matrix irreducible?}$$

Answer: All the states do not communicate with each other. The chain is not irreducible and not ergodic. Because the state 3 is an absorbing state.

Problem: Find the nature of states of the Markov chain with transition probability matrix

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

Answer: The Markov chain is finite and irreducible; all its states are non-null persistent. All states are not ergodic.

STEADY STATE CONDITION:

Stable Probability: In many Markov chains, the probability for a particular state will approach a limiting value as time goes to infinity. In other words, in the far future, the probability won't be changing much from one transition to the next transition. These limiting values are called the stable probabilities.

Steady State Condition: If a system is such that each state has probability equal to its stable probability, the probability will persist for all the time. Then the system is said to be in steady state condition.

STEADY STATE VECTOR (OR) EQUILIBRIUM VECTOR OF A MARKOV CHAIN

If a Markov chain with transition matrix P is regular, then there is a unique vector V such that, for any probability vector V and for large values of n , $V \cdot P^n = V$. Vector V is called the **equilibrium vector** or the **fixed vector** or **steady state vector** of the Markov chain. This is also called long range trend of the Markov Chain.

Probability Vector: A probability vector is a matrix of only one row, having non-negative entries, with the sum of the entries equal to 1. i.e., $v_1 + v_2 + v_3 + \dots + v_n = 1$ for n states.

Theorem: If a Markov chain with transition matrix P is regular, then there exists a probability vector V such that $V \cdot P = V$

Proof: By the definition of equilibrium vector, we have

$$V \cdot P^n \cong V \quad \text{Multiplying with } P \text{ on both sides,}$$

$$V \cdot P^n \cdot P \cong V \cdot P$$

$$\Rightarrow V \cdot P^{n+1} \cong V \cdot P \quad \dots\dots (1)$$

Since $V \cdot P^n \cong V$ for large volume of n , it is also true that $V \cdot P^{n+1} \cong V$ for large values of n .

$$\text{Thus, we have } V \cdot P^{n+1} \cong V \quad \dots\dots (2)$$

From (1) and (2), we get $V \cdot P = V$

This vector V gives the long-range trend of the Markov chain.

Example: A house wife buys 3 kinds of cereals A, B and C. She never buys the same cereals in successive weeks. If she buys Cereal A, the next week she buys B. However, if she buys B or C, the next week it is 3 times as likely to buy A as other cereals. In the long run, how often does she buy each of the three cereals.

Solution:

Step 1: Define the states

Let

- A = buys cereal A
- B = buys cereal B
- C = buys cereal C

She **never buys the same cereal in successive weeks**, so

$$P(A \rightarrow A) = P(B \rightarrow B) = P(C \rightarrow C) = 0.$$

Step 2: Construct the transition probability matrix

From A

If she buys A, the next week she **must buy B**.

Therefore,

$$P(A \rightarrow B) = 1, \quad P(A \rightarrow C) = 0.$$

So the first row is

$$[0, 1, 0].$$

From B

If she buys B, the next week A is **3 times as likely as each of the other cereals**.

She cannot buy B again. Therefore, only A and C are possible.

Let

$$P(B \rightarrow C) = x.$$

Then

$$P(B \rightarrow A) = 3x.$$

Since the probabilities must add to 1,

$$3x + x = 1$$

$$4x = 1$$

$$x = \frac{1}{4}.$$

Hence,

$$P(B \rightarrow A) = \frac{3}{4}, \quad P(B \rightarrow C) = \frac{1}{4}.$$

The second row is

$$\left[\frac{3}{4}, 0, \frac{1}{4} \right].$$

From C

Similarly, if she buys C, she cannot buy C next week, and A is 3 times as likely as B.

Thus,

$$P(C \rightarrow A) = \frac{3}{4}, \quad P(C \rightarrow B) = \frac{1}{4}.$$

The third row is

$$\left[\frac{3}{4}, \frac{1}{4}, 0 \right].$$

Therefore, the transition probability matrix is

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{4} & 0 & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{4} & 0 \end{bmatrix}$$

Step 3: Find the steady-state probabilities



Let the long-run probabilities of buying A, B and C be

$$\pi_A, \quad \pi_B, \quad \pi_C.$$

The equilibrium condition is

$$\pi P = \pi$$

where

$$\pi = (\pi_A, \pi_B, \pi_C).$$

Thus,

$$(\pi_A, \pi_B, \pi_C) \begin{bmatrix} 0 & 1 & 0 \\ \frac{3}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix} = (\pi_A, \pi_B, \pi_C).$$

This gives the equations:

$$\pi_A = \frac{3}{4}\pi_B + \frac{3}{4}\pi_C \quad (1)$$

$$\pi_B = \pi_A + \frac{1}{4}\pi_C \quad (2)$$

$$\pi_C = \frac{1}{4}\pi_B \quad (3)$$

and

$$\pi_A + \pi_B + \pi_C = 1. \quad (4)$$

From (3),

$$\pi_C = \frac{1}{4}\pi_B.$$

Substitute this into (2):

$$\pi_B = \pi_A + \frac{1}{4} \left(\frac{1}{4}\pi_B \right)$$

$$\pi_B = \pi_A + \frac{1}{16}\pi_B$$

$$\pi_A = \frac{15}{16}\pi_B.$$

Also,

$$\pi_C = \frac{1}{4}\pi_B.$$

Now use

$$\pi_A + \pi_B + \pi_C = 1.$$

Therefore,

$$\frac{15}{16}\pi_B + \pi_B + \frac{1}{4}\pi_B = 1.$$

Taking denominator 16,

$$\frac{15 + 16 + 4}{16}\pi_B = 1$$

$$\frac{35}{16}\pi_B = 1$$

$$\pi_B = \frac{16}{35}.$$

Hence,

$$\pi_A = \frac{15}{16} \times \frac{16}{35} = \frac{15}{35} = \frac{3}{7}$$

and

$$\pi_C = \frac{1}{4} \times \frac{16}{35} = \frac{4}{35}.$$

Final Answer

$$\pi_A = \frac{15}{35}, \quad \pi_B = \frac{16}{35}, \quad \pi_C = \frac{4}{35}$$

or approximately,

Cereal	Long-run proportion	Percentage
A	15/35=3/7	42.86%
B	16/3516/35	45.71%
C	4/354/35	11.43%

Therefore, in the long run, she buys A about 42.86% of the weeks, B about 45.71% of the weeks, and C about 11.43% of the weeks.

Problem: The weather in certain spot is classified as fair, cloudy (without rain) or rainy. A fair day is followed by a fair day 60% of the time and by a cloudy day 25% of the time. A cloudy day is followed by a cloudy day 35% of the time and by a rainy day 25% of the time. A rainy day is followed by a cloudy day 40% of the time and by a rainy day 25% of the time. Initial probabilities are 0.3, 0.3 and 0.4. Find the probability that there will be rainy day after 3 days. What portion of the days is expected to be fair, cloudy or rainy in the long run.

1. Define the states

Let

- F = Fair
- C = Cloudy (without rain)
- R = Rainy

The given probabilities are:

- Fair $\rightarrow$ Fair = 0.60, Fair $\rightarrow$ Cloudy = 0.25
- Cloudy $\rightarrow$ Cloudy = 0.35, Cloudy $\rightarrow$ Rainy = 0.25
- Rainy $\rightarrow$ Cloudy = 0.40, Rainy $\rightarrow$ Rainy = 0.25

Since the probabilities in each row must sum to 1, the missing probabilities are:

$$P(F \rightarrow R) = 1 - 0.60 - 0.25 = 0.15$$

$$P(C \rightarrow F) = 1 - 0.35 - 0.25 = 0.40$$

$$P(R \rightarrow F) = 1 - 0.40 - 0.25 = 0.35$$

Therefore, the transition probability matrix is

$$P = \begin{bmatrix} 0.60 & 0.25 & 0.15 \\ 0.40 & 0.35 & 0.25 \\ 0.35 & 0.40 & 0.25 \end{bmatrix}$$

with the order F, C, R .

The initial probability vector is

$$P^{(0)} = (0.3, 0.3, 0.4).$$

Part I: Probability of a rainy day after 3 days

We need

$$P(X_3 = R).$$

We calculate successively:

$$P^{(1)} = P^{(0)} P$$

After 1 day

$$(0.3, 0.3, 0.4) \begin{bmatrix} 0.60 & 0.25 & 0.15 \\ 0.40 & 0.35 & 0.25 \\ 0.35 & 0.40 & 0.25 \end{bmatrix}$$

Therefore,

$$P^{(1)} = (0.18 + 0.12 + 0.14, 0.075 + 0.105 + 0.16, 0.045 + 0.075 + 0.10)$$

$$P^{(1)} = (0.44, 0.34, 0.22)$$

After 2 days

$$\begin{aligned}P^{(2)} &= P^{(1)}P \\ &= (0.44, 0.34, 0.22)P\end{aligned}$$

For the three states:

$$\begin{aligned}P(F) &= 0.44(0.60) + 0.34(0.40) + 0.22(0.35) \\ &= 0.264 + 0.136 + 0.077 = 0.477\end{aligned}$$

Similarly,

$$\begin{aligned}P(C) &= 0.44(0.25) + 0.34(0.35) + 0.22(0.40) \\ &= 0.11 + 0.119 + 0.088 = 0.317\end{aligned}$$

and

$$\begin{aligned}P(R) &= 0.44(0.15) + 0.34(0.25) + 0.22(0.25) \\ &= 0.066 + 0.085 + 0.055 = 0.206.\end{aligned}$$

Thus,

$$P^{(2)} = (0.477, 0.317, 0.206)$$

After 3 days

$$P^{(3)} = P^{(2)}P$$

We are particularly interested in the rainy probability:

$$\begin{aligned}P(X_3 = R) &= 0.477(0.15) + 0.317(0.25) + 0.206(0.25) \\ &= 0.07155 + 0.07925 + 0.05150\end{aligned}$$

$$P(X_3 = R) = 0.2023$$

Therefore,

$$\text{Probability of a rainy day after 3 days} = 0.2023$$

or approximately

$$20.23\%.$$

For completeness,

$$P^{(3)} = (0.48615, 0.31155, 0.20230).$$

Part II: Long-run proportion of fair, cloudy and rainy days

For the long-run probabilities, let

$$\boldsymbol{\pi} = (\pi_F, \pi_C, \pi_R).$$

At equilibrium,

$$\boldsymbol{\pi}P = \boldsymbol{\pi}$$

and

$$\pi_F + \pi_C + \pi_R = 1.$$

Thus,

$$(\pi_F, \pi_C, \pi_R) \begin{bmatrix} 0.60 & 0.25 & 0.15 \\ 0.40 & 0.35 & 0.25 \\ 0.35 & 0.40 & 0.25 \end{bmatrix} = (\pi_F, \pi_C, \pi_R).$$

This gives

$$\pi_F = 0.60\pi_F + 0.40\pi_C + 0.35\pi_R \quad (1)$$

$$\pi_C = 0.25\pi_F + 0.35\pi_C + 0.40\pi_R \quad (2)$$

$$\pi_R = 0.15\pi_F + 0.25\pi_C + 0.25\pi_R. \quad (3)$$

Using equations (1), (2), and

$$\pi_F + \pi_C + \pi_R = 1,$$

we obtain

$$\pi_F \approx 0.4865$$

$$\pi_C \approx 0.3119$$

$$\pi_R \approx 0.2016.$$

Weather	Long-run proportion	Percentage
Fair	0.48650.4865	48.65%
Cloudy	0.31190.3119	31.19%
Rainy	0.20160.2016	20.16%

So, in the long run, approximately:

$$\text{Fair: 48.65\%, Cloudy: 31.19\%, Rainy: 20.16\%}$$

Notice that the initial probabilities $(0.3, 0.3, 0.4)$ are used to find the probability after 3 days, whereas the steady-state probabilities are obtained from $\boldsymbol{\pi}P = \boldsymbol{\pi}$ and describe the long-run behavior.